

Legalease: AI-Powered Legal Document Generator

Complete Implementation Source Code Architecture (Python / Flask / OpenAI API)

1. Project Overview

Legalease is a comprehensive web application engineered to automate the generation, structuring, and management of legally binding agreements. Powered by the OpenAI API and built on a lightweight Flask backend, the architecture handles dynamic user inputs, structures prompt legal payloads, renders clean text layers, and outputs finalized PDF instruments securely.

2. Backend Core API (app.py)

```
# app.py
import os
from flask import Flask, render_template, request, jsonify, send_file
import openai
from dotenv import load_dotenv
from fpdf import FPDF

load_dotenv()
app = Flask(__name__)
openai.api_key = os.getenv("OPENAI_API_KEY")

@app.route('/')
def index():
    return render_template('index.html')

@app.route('/generate', methods=['POST'])
def generate_document():
    data = request.json
    doc_type = data.get('doc_type')
    party_a = data.get('party_a')
    party_b = data.get('party_b')
    jurisdiction = data.get('jurisdiction')
    key_clauses = data.get('key_clauses', '')

    prompt = f"Write a professional, legally binding {doc_type} governed by the laws of {jurisdiction}.\n" \
            f"Party A: {party_a}\nParty B: {party_b}\n" \
            f"Include the following specific requirements/clauses: {key_clauses}.\n" \
            f"Ensure standard legal definitions, confidentiality, dispute resolution, and execution blocks are included."

    try:
        response = openai.ChatCompletion.create(
            model="gpt-4",
```

```

        messages=[
            {"role": "system", "content": "You are an expert AI legal counsel
drafting precise, industry-standard legal documents."},
            {"role": "user", "content": prompt}
        ],
        temperature=0.3
    )
    legal_text = response.choices[0].message['content']
    return jsonify({"success": True, "document": legal_text})
except Exception as e:
    return jsonify({"success": False, "error": str(e)}), 500

@app.route('/download', methods=['POST'])
def download_pdf():
    data = request.json
    content = data.get('content', '')

    pdf = FPDF()
    pdf.add_page()
    pdf.set_font("Arial", size=11)

    # Simple multi-line text processing
    for line in content.split('\n'):
        pdf.multi_cell(0, 5, txt=line.encode('latin-1', 'replace').decode('latin-1'))

    output_path = "temp_legal_doc.pdf"
    pdf.output(output_path)
    return send_file(output_path, as_attachment=True, download_name="legal_agreement.pdf")

if __name__ == '__main__':
    app.run(debug=True)

```

3. User Interface (templates/index.html)

```

<!-- templates/index.html --><br/><!DOCTYPE html><br/><html lang="en"><br/><head><br/>
    <meta charset="UTF-8"><br/>    <title>Legalease - AI Legal Document Generator</title><br/>
    <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/tailwindcss/2.2.19/tailwind.min.css"><br/></head><br/><body class="bg-gray-50 font-sans"><br/>
    <div class="max-w-4xl mx-auto py-12 px-4"><br/>        <h1 class="text-3xl font-bold text-gray-900 mb-2">Legalease AI</h1><br/>
        <p class="text-gray-600 mb-8">Generate enterprise-grade, custom legal agreements instantly.</p><br/>
        <div class="grid grid-cols-1 md:grid-cols-2 gap-8"><br/>
            <!-- Form Block --><br/>
            <div class="bg-white p-6 rounded-lg shadow-sm border border-gray-200"><br/>
                <div class="mb-4"><br/>
                    <label class="block text-sm font-semibold text-gray-700 mb-1">Document Type</label><br/>
                    <select id="doc_type" class="w-full border p-2 rounded"><br/>
                        <option value="Non-Disclosure Agreement (NDA)">Non-Disclosure Agreement (NDA)</option><br/>
                        <option value="Independent Contractor Agreement">Independent Contractor Agreement</option><br/>
                        <option value="Software License Agreement">Software License Agreement</option><br/>
                    </select><br/>
                </div><br/>
                <div class="mb-4"><br/>
                    <label class="block text-sm font-semibold text-gray-700 mb-1">Party A (First Party)</label><br/>
                    <input type="text" id="party_a" class="w-full bord

```

```

er p-2 rounded" placeholder="Company / Full Name"><br/>                                </div><br/>
    <div class="mb-4"><br/>                                <label class="block text-sm font-semibold text-gray-700 mb-1">Party B (Second Party)</label><br/>
    <input type="text" id="party_b" class="w-full border p-2 rounded" placeholder="Individual / Entity"><br/>                                </div><br/>                                <div
    class="mb-4"><br/>                                <label class="block text-sm font-semibold text-gray-700 mb-1">Jurisdiction / State Law</label><br/>                                <input
    type="text" id="jurisdiction" class="w-full border p-2 rounded" placeholder="e.g. , California, Delaware"><br/>                                </div><br/>                                <div class
    ="mb-4"><br/>                                <label class="block text-sm font-semibold text-gray-700 mb-1">Custom Clauses / Provisions</label><br/>                                <textarea
    id="key_clauses" rows="4" class="w-full border p-2 rounded" placeholder="Add custom terms..."></textarea><br/>                                </div><br/>                                <button on
    click="generateDoc()" class="w-full bg-blue-600 text-white py-2 rounded font-medium hover:bg-blue-700">Generate Agreement</button><br/>                                </div><br/><br/>
    <!-- Preview Block --><br/>                                <div class="bg-white p-6 rounded
    -lg shadow-sm border border-gray-200 flex flex-col"><br/>                                <h3 class
    ="text-lg font-semibold mb-2">Draft Preview</h3><br/>                                <textarea id=
    "output" readonly class="w-full flex-1 border p-3 rounded font-mono text-sm bg-gray-50 mb-4 resize-none" placeholder="Your generated contract will appear here...">
    </textarea><br/>                                <button onclick="downloadPDF()" id="downloadBtn" di
    sabled class="w-full bg-green-600 text-white py-2 rounded font-medium hover:bg-green-700 opacity-50 cursor-not-allowed">Download Document PDF</button><br/>
    </div><br/>                                </div><br/>                                </div><br/><br/>                                <script><br/>                                let
    fullDraftText = "";<br/>                                async function generateDoc() {<br/>                                con
    st payload = {<br/>                                doc_type: document.getElementById('doc_type').v
    alue,<br/>                                party_a: document.getElementById('party_a').value,<br/>
    party_b: document.getElementById('party_b').value,<br/>
    jurisdiction: document.getElementById('jurisdiction').value,<br/>
    key_clauses: document.getElementById('key_clauses').value<br/>                                };<br
    />                                const res = await fetch('/generate', {<br/>                                method: '
    POST',<br/>                                headers: {'Content-Type': 'application/json'},<br/>
    body: JSON.stringify(payload)<br/>                                });<br/>                                cons
    t data = await res.json();<br/>                                if (data.success) {<br/>
    fullDraftText = data.document;<br/>                                document.getElementById('outpu
    t').value = fullDraftText;<br/>                                document.getElementById('downloadBt
    n').disabled = false;<br/>                                document.getElementById('downloadBtn').c
    lassList.remove('opacity-50', 'cursor-not-allowed');<br/>                                } else {<br/>
    alert("Error generating document: " + data.error);<br/>
    }<br/>                                }<br/><br/>                                async function downloadPDF() {<br/>                                c
    onst res = await fetch('/download', {<br/>                                method: 'POST',<br/>
    headers: {'Content-Type': 'application/json'},<br/>                                bod
    y: JSON.stringify({content: fullDraftText})<br/>                                });<br/>                                co
    nst blob = await res.blob();<br/>                                const link = document.createElement('
    a');<br/>                                link.href = window.URL.createObjectURL(blob);<br/>
    link.download = "legal_agreement.pdf";<br/>                                link.click();<br/>
    }<br/>                                </script><br/></body><br/></html>

```

4. Environment Configuration (.env)

```

# .env
OPENAI_API_KEY=your_secured_openai_api_key_here
FLASK_APP=app.py
FLASK_ENV=development

```

5. Python Dependencies (requirements.txt)

```
flask==3.0.0  
openai==0.28.1  
python-dotenv==1.0.0  
fpdf==1.7.2
```

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